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GIXRD AND XPS INVESTIGATION OF SILICIDATION IN ION BEAM MIXED Cu/Si(1 1 1) SYSTEM

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Grazing incidence X-ray diffraction (GIXRD) and X-ray photoelectron spectroscopy (XPS) are used for phase identification and the determination of chemical states of the Cu/Si(1 1 1) system ion beam mixed at room temperature (RT) and 100°C. Cu 2*p* and Si 2*p* photoelectron peaks are analyzed to study the silicide formation. GIXRD study shows the formation of silicide phases in samples irradiated both at RT and 100°C. © 1998 Elsevier Science Ltd. All rights reserved

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Copper has been proposed as a new material for metalization in Si-based semiconductor devices [1] due to its low resistivity ($\rho < 2 \mu\Omega \text{ cm}^{-1}$) and excellent electromigration resistance. Compared to other noble metals copper is one of the fastest diffusing elements in Si, diffusing interstitially with an activation energy of 0.43 eV [2]. Additionally, Cu tends to oxidize in air at relatively low temperatures $\sim 250^\circ\text{C}$, giving rise to a substantial increase in resistivity [3]. The copper silicides are reported to form at low temperature at the interface of Cu thin films deposited on Si substrates [4]. A study of the formation of copper silicide during thermal annealing, their oxidation behaviour and electronic and electrical properties have been reported by Cros *et al.* [5]. Though the ion beam mixing in noble metal–Si system has been studied extensively, most of the work has been carried out in Au–Si system [6, 7]. There are very few studies on the ion beam mixing and consequent phase formation in Cu–Si system [8].

In this paper we report for the first time GIXRD and XPS investigations of silicidation in ion beam mixed Cu/Si(1 1 1) system.

The Cu films of 500 Å thickness was deposited on cleaned Si(1 1 1) by thermal evaporation at

8×10^{-6} mbar vacuum at the rate of 2 Å per second. The samples, thus prepared were irradiated with 80 keV Ar⁺ ions to a dose of 1×10^{16} ions cm⁻² using a low energy accelerator at room temperature (RT) and 100°C. The substrate temperature during irradiation was chosen to be below the temperature required for Cu–Si interfacial reaction ($\sim 200^\circ\text{C}$) [5, 9]. The energy of the argon ions (80 keV) was chosen so that the projected range of the ions match with the position of the Cu/Si interface. GIXRD was performed on a Siemens Diffractometer using Cu-*K_α* radiation. The incident angle used in the study was 0.6°. The diffraction data was recorded for a 2θ range 30° to 60°. XPS studies were performed at a vacuum of 1.7×10^{-10} mbar using a hemispherical analyzer at 20 eV pass energy [6]. *In-situ* depth profiling during XPS investigations was carried out using Ar ion beam of energy 4 keV and beam current 1 μA.

Figure 1(a) shows the GIXRD pattern of the as-deposited Cu/Si(1 1 1) sample. Cu(1 1 1) is the prominent peak along with the Cu(2 0 0). The pattern is consistent with the Joint Committee for Powder Diffraction Society (JCPDS) file [10]. Figure 1(b) shows the GIXRD pattern of the sample irradiated with Ar ions at RT. The GIXRD indicates the formation of copper silicides. Several phases of copper silicide (Cu₃Si, Cu₄Si, Cu₅Si, Cu₁₅Si₄ and Cu_{0.83}Si_{0.17}) [11] with closely matching *d* spacing

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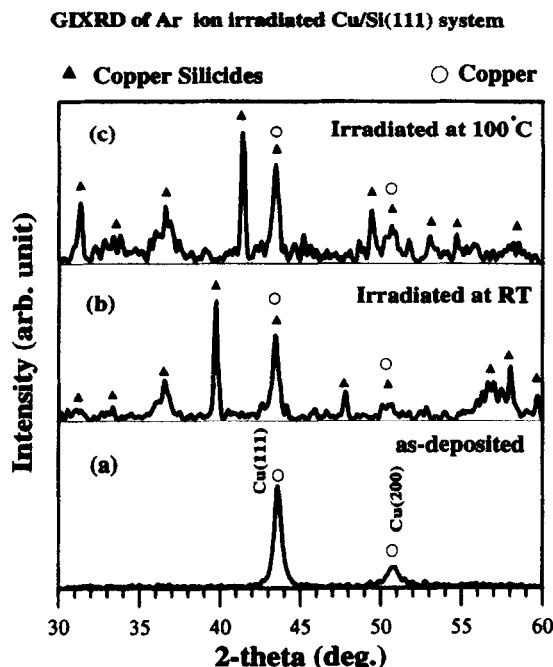


Fig. 1. Grazing incidence X-ray diffraction (GIXRD) pattern of (a) as-deposited Cu/Si(1 1 1) sample, (b) sample irradiated with 80 keV Ar ions at room temperature to a dose of 1×10^{16} ions cm^{-2} and (c) sample irradiated at 100°C (same energy and dose).

and with slight difference in stoichiometry exist. Hence it is difficult to uniquely identify the silicide phase which has formed during mixing. The possibility of the multiple phases can not be ruled out. Dhar *et al.* [12] reported the possibility of the presence of Cu_3Ge and $\text{Cu}_{0.83}\text{Ge}_{0.17}$ phases along with unreacted Cu and Ge in the case of ion beam mixing of Cu–Ge multilayers. The Fig. 1(c) shows the GIXRD pattern of the sample irradiated at 100°C. Most of the peaks associated with silicides appear in this diffraction pattern also. The minor differences in the GIXRD pattern of the RT and 100°C irradiated samples may be attributed to the preferred orientational growth of silicide.

The XPS spectra ($\text{Cu } 2p_{3/2}$ photoelectric peak) of the as deposited Cu/Si(1 1 1) sample as well as the sample subjected to ion beam mixing at room temperature are shown in Fig. 2. The $\text{Cu } 2p_{3/2}$ peak for the as-deposited case is seen at 932.5 eV with a FWHM of 1.5 eV [Fig. 2(a)]. Figure 2(b) represent the $\text{Cu } 2p_{3/2}$ peak for the Cu/Si(1 1 1) sample irradiated at room temperature (RT). The peak is very broad probably due to the presence of several phases of Cu at the surface including surface contamination. Figure 2(c) and 2(d) are the XPS spectra for the same sample after *in-situ* sputtering of the surface contamination layer for 300 s and 2100 s, respectively. The $\text{Cu } 2p_{3/2}$ peak has sharpened due to the removal of the surface contamination. FWHM of

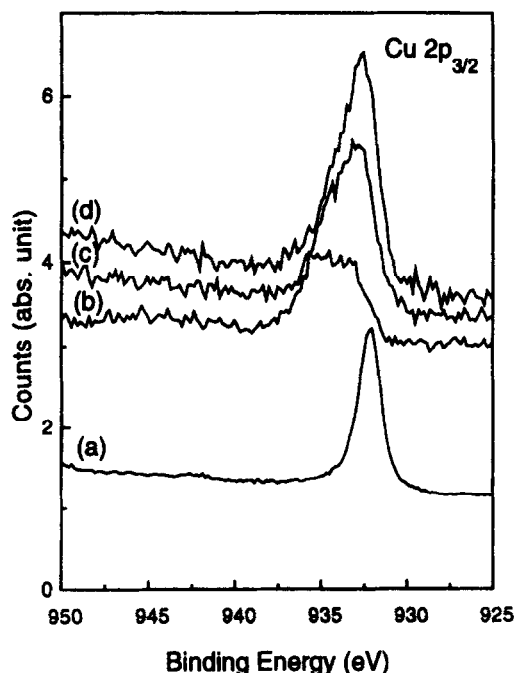


Fig. 2. Cu $2p_{3/2}$ photoelectron spectra of Cu/Si(1 1 1) (a) as-deposited; sample irradiated with 80 keV Ar ions at room temperature to a dose of 1×10^{16} ions cm^{-2} , (b) on the surface, (c) after sputtering for 300 s and (d) after sputtering for 2100 s.

Cu $2p_{3/2}$ in Fig. 2(d) is 2.2 eV and the binding energy is 932.8 eV. The position of Cu $2p_{3/2}$ peak in the pure Cu and the Cu in the silicide phase are very close as reported by Cros *et al.* [5]. However in our observation we have observed the a peak shift of 0.3 eV and a broadening by 0.7 eV as compared to pure Cu. This peak shift is very small and within the energy resolution of the XPS instrument (0.4 eV). However the peak shift might still be attributed to silicide formation considering that we find evidence for silicidation from GIXRD study.

Figure 3 shows the XPS spectra ($\text{Cu } 2p_{3/2}$) of the sample ion beam mixed at 100°C along with that of as-deposited sample. The surface shows [Fig. 3(b)] the presence of contamination. But after sputtering for 300 s the surface does not show the presence of any contamination [Fig. 3(c)]. The $\text{Cu } 2p_{3/2}$ peak appears at 932.8 with FWHM of 2.2 eV corresponding to the silicide phase as mentioned previously in the case of sample irradiated at RT [Fig. 2(d)].

The XPS spectra of Si $2p$ are shown in Fig. 4(c,d) for the sample irradiated at RT and 100°C. These spectra were recorded after sputtering the surface for 1500 s. The Si $2p$ line has split into two distinct peaks. The peak around 99.0 eV is the characteristic peak of pure silicon as well as the silicon in copper silicides [5, 16] and the peak around 103.5 eV is the characteristic peak of Si-oxide (SiO_x , $x < 2$). The Si $2p$ peak for pure silicon

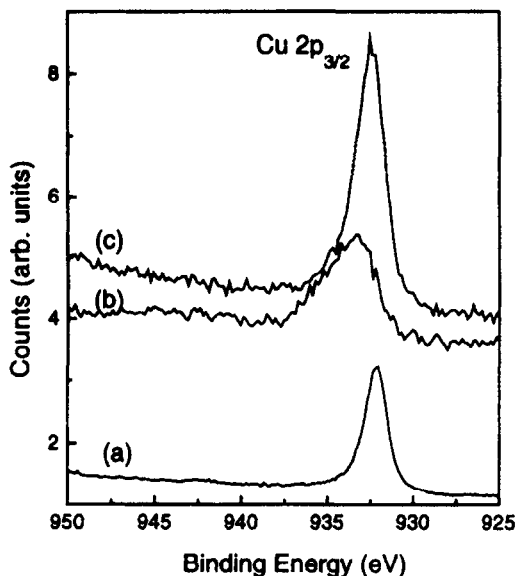


Fig. 3. Cu $2p_{3/2}$ photoelectron spectra for Cu/Si(1 1 1) system (a) as-deposited; sample irradiated with 80 keV Ar ions at 100°C to a dose of 1×10^{16} ions cm^{-2} , (b) on the surface, (c) after sputtering 300 s.

and silicon in copper silicide are indistinguishable [5]. Our depth profile study shows the peak at 103.5 eV is present throughout but is slowly reducing in intensity with sputtering time which may be attributed to the reduction in the amount of Si-oxide with depth. The O 1s peak shown in Fig. 4(a) and (b) are observed at

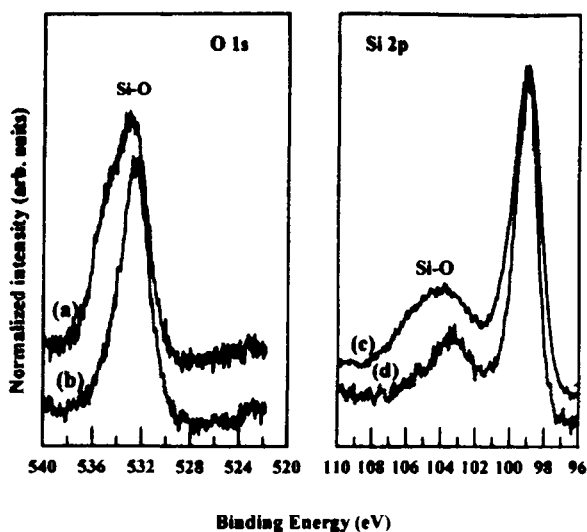


Fig. 4. Photoelectron spectra of Ar ion irradiated Cu/Si(1 1 1) system after sputtering 1500 s from the surface; (a) and (b) gives the O 1s photoelectric peak for the sample irradiated at RT and the sample irradiated at 100°C respectively; (c) and (d) gives the Si 2p photoelectric peak for the sample irradiated at RT and the sample irradiated at 100°C respectively.

around 532.5 eV is due to the formation of Si-oxide which is also reflected to the Si 2p photoelectron peak. There is no substantial oxidation of copper as evidenced by the Cu $2p_{3/2}$ peak shape and the O 1s binding energy. Presence of CuO is always associated with a chemical shift of the Cu $2p_{3/2}$ peak by 1.2 eV and a satellite peak situated at 10 eV higher binding energy [5]. There is no satellite peak or energy shift indicating the formation of CuO. The presence of Cu₂O is indistinguishable from pure copper because the binding energy of Cu $2p_{3/2}$ in Cu₂O and pure Cu are very close to each other [5]. In the present case the Cu 2p photoelectron peaks are observed to be broad at the surface as shown in Fig. 2(b) and Fig. 3(b) which may be due to the trace of Cu₂O on the surface. The films are deposited in a separate chamber at high- 10^{-6} mbar vacuum, so oxygen is present only from residual contamination.

The driving force for such preferential oxidation of Si is the large free energy of formation of SiO₂ ($-856.3 \text{ kJ mol}^{-1}$) as compared to CuO (-129 kJ mol^{-1}) or Cu₂O (-146 kJ mol^{-1}). Similar oxidation has been observed in the case of Au₄Si alloys [13], Pd₄Si alloys, Ni₄Si silicides [14] when studied under controlled oxidation conditions.

In conclusion the Cu/Si(1 1 1) ion beam mixed at RT as well as 100°C shows the formation of copper silicide as confirmed by GIXRD. The observed chemical shift of Cu $2p_{3/2}$ due to silicidation is small (0.3 eV) within the energy resolution of the XPS equipment (0.4 eV). The trace of Si-oxide is observed along with the Cu-silicide.

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